YIORGOS PAPADOPOULOS

Thessaloniki, Greece • giorgospapadopoulos002@yahoo.com
linkedin.com/in/yiorgospapadopoulos/ • github.com/yiorgospapadopoulos

ABOUT ME

Computer Science graduate and MSc student in Artificial Intelligence and Applications at the University of Thessaly, building production Python backends that integrate ML and NLP capabilities. Experienced in training and benchmarking models, designing typed services, and delivering APIs from design through code review to deployment.

PROFESSIONAL EXPERIENCE

Machine Learning and LLM Systems Engineer | Zimmerman February 2026 – Present

- Building backend services that integrate ML, NLP, LLMs, and embedding-based retrieval, exposed through typed APIs and React frontends.
- Owning backend features end-to-end (see Projects), from design through PR review to deployment.
- Working closely with senior engineers through multi-round code reviews on GitHub, raising the bar on code quality, typing discipline, and architectural clarity.
- Operating in an agile workflow (Jira sprints, shippable tickets) and presenting completed work to the engineering team.

EDUCATION

MSc in Artificial Intelligence and Applications, University of Thessaly October 2026 – Present

BSc in Computer Science & Telecommunications, University of Thessaly 2021 – 2026

- GPA: 7.5/10
- Specialization in Data Management and Artificial Intelligence
- Thesis focused on the development of artificial intelligence models

Erasmus+ International Exchange | ISLA Polytechnic Institute, Porto 2024 – 2025

- Completed an international study program in Portugal during the 2024–2025 academic year.

CERTIFICATIONS

- AI in Healthcare Specialization, Stanford University. 5-course specialization covering clinical machine learning implementation, algorithmic fairness, and healthcare regulatory frameworks.
- Generative AI for Healthcare, Google Cloud (2026).

TECHNICAL PROJECTS

Thesis: Multi-Model Deep Learning for Chest X-Ray Pathology Classification *(Graded 10/10)*

- Designed and trained a fusion architecture combining MobileNetV2 and ResNet50 backbones for multi-label pathology classification on the NIH ChestX-ray14 dataset (~112K images, 14 labels), with all models achieving 90%+ AUC.
- Implemented Grad-CAM visualization to produce per-class saliency maps, making model predictions interpretable for clinical review.
- Built a PyQt5 desktop application wrapping the trained model, enabling end-to-end inference and visualization from a single interface.

Model Pipeline: ML Service for Grant-Management Analytics *(Zimmerman)*

- Audited and corrected ML models across 24 use cases (regression, classification, clustering, anomaly detection, optimization, network analysis, causal inference) in a Python/Flask service: removed target leakage, fixed data aggregation and train/test splits, aligned prediction inputs with production data, and capped model size to cut saved artifacts from several GB to ~1MB.
- Added experiment tracking (per-run logging of config, winning model, hyperparameters, data and git versions), deterministic run replay, and a data migration script.
- Drove the service to 100% test coverage (1,300+ tests) behind a pre-commit gate of ruff, flake8, bandit, and type checking, over Pydantic-typed config and a Swagger-documented REST API.

AI Lead Generation Agent: Automated Sourcing from IATI Funding Data *(Zimmerman)*

- Built a Python/Flask pipeline querying the IATI Datastore (Solr, ~875K activities) to source and qualify sales leads from international aid funding data, replacing a manual process.
- Implemented a signal-detection layer of five org-level detectors (new funding, first-time publisher, activity spike, dormant publisher, large commitment), each running scoped Solr queries and returning structured evidence for review.
- Built a three-stage contact-enrichment module (cached prior runs, then IATI data, then OpenAI web search as fallback) that cut paid LLM calls by reusing earlier results, with per-run token and cost tracking.
- Ran a month-by-month historical replay to measure lead volume, filtering out existing clients, multilaterals, and large publishers; typed, class-based Python at 100% test coverage enforced via pre-commit.

AIDA: Semantic Search Platform for IATI Aid Data *(Zimmerman)*

- Built a Python/Flask backend exposing REST endpoints for semantic search over IATI international aid data, consumed by a React frontend.
- Designed and implemented multiple search strategy modules (fuzzy string matching, dense embedding retrieval, and a hybrid LLM-assisted approach) and benchmarked them on accuracy, latency, and cost.
- Integrated NLP/embedding models into the search pipeline, with structured logging, typed Python, class-based design, and clear separation of concerns across retrieval, ranking, and API layers.
- Worked through multi-round PR reviews on GitHub (rebasing, squashing, addressing detailed senior feedback) and contributed via Jira-tracked sprints; presented the system internally to the engineering team.

PROGRAMMING SKILLS

- **Libraries & Frameworks:** TensorFlow, PyTorch, Scikit-learn, Flask, Pandas, OpenCV
- **Languages:** Python, HTML, C++, JavaScript, SQL
- **Databases:** MySQL
- **Tools & Practices:** Git/GitHub, code review, Jira, Docker, agile workflows
- **Collaboration:** Experienced working in distributed engineering teams, navigating code review feedback, and aligning on technical direction with senior developers.

LANGUAGES AND INTERESTS

- **English:** ECPE (C2 – Proficiency, With Honors)
- **Greek:** Native
- **Interests:** Strong interest in fashion and visual arts, incorporating creativity into technical projects.